Online Supplement for "Conservative Test-Time Adaptation under Open-World Contamination via Binary Correctness Feedback"

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1 Additional methodological notes

Binary-feedback accounting. The main method uses paired binary correctness evaluation on queried examples. For a proposed candidate update, binary correctness is computed before and after the update on the same queried subset, and the candidate is accepted only if the empirical paired gain exceeds threshold. This is why the PPSN version uses the term *binary correctness feedback* rather than a stronger "one-bit" slogan.

Delayed commitment. The current lag variant evaluates the paired gate immediately and delays applying an accepted candidate for several windows. It should therefore be interpreted as delayed commitment of an already scored candidate, not as a full simulation of labels arriving several windows later.

2 Baseline porting disclosure

Several non-core baselines were ported into a unified local CIFAR streaming harness so that all methods share the same source checkpoints, stream construction, query-rate grid, and reporting rules. These comparisons are intended to isolate behavior under a common protocol, not to claim exact end-to-end reproduction of each official runner.

Additional porting notes are provided in `docs/baseline_porting_notes.md`.

3 Supervision-accounting tables

4 Full benchmark tables

5 Reproducibility companion

Implementation details, resolved configurations, result files, and repository-level notes are documented in the companion files:

Table 1. Shared-protocol porting disclosure for non-core baselines.

Baseline	Official idea retained	Unified-harness changes	Why the comparison remains informative
BiTTA-style	Queried binary feedback drives positive and negative adaptation decisions.	Implemented inside the shared CIFAR streaming harness with the same source checkpoints, windows, query-rate grid, and reporting pipeline; no separate official memory runner is invoked.	Tests whether binary feedback remains useful under the same backbone, stream, and budget protocol as the proposed gate.
ATTA-style	Budgeted queried labels are used during test-time adaptation.	Ported to the shared harness with the same stream API, query grid, and evaluation metrics rather than an external active-learning stack.	Provides a stronger queried-label reference under the same target streams and query accounting.
EATTA-style	Low-label updates are combined with reliable unlabeled samples.	Reliability heuristics are adapted to the local harness while keeping the same backbone family, windows, and query budgets.	Clarifies what richer low-label supervision can achieve under the same deployment protocol.
COME-style	Conservative label-free entropy/stability control is retained as a 0-bit baseline.	Implemented in the same batch-normalization adaptation harness and source-model family as Tent and the binary gate.	Gives a matched conservative 0-bit comparator under the same online adaptation pipeline.

Table 2. Family-level supervision and performance summary at the primary operating point.

Method	Supervision	Events / 1k	C10 frontier gain	C10 contam gain	C10 contam NTR	C100 frontier gain	C100 contam gain	C100 contam NTR
Frozen	0-bit	0.0	0.000	0.000	0.000	0.000	0.000	0.000
Tent	0-bit	0.0	0.151	0.045	0.133	0.179	0.073	0.067
FB-Gate-Tent	binary correctness events	20.0	0.147	0.029	0.000	0.156	0.047	0.000
Self-Gate Tent (0-bit)	0-bit	0.0	0.149	0.048	0.100	0.167	0.069	0.033
Throttle Tent (0-bit)	0-bit	0.0	0.094	0.045	0.133	0.179	0.073	0.067
COME-style 0-bit	0-bit	0.0	0.150	0.049	0.133	0.178	0.077	0.033
BiTTA-style	binary correctness events	10.0	0.090	0.007	0.400	0.101	0.026	0.167
ATTA-style	label queries	10.0	0.071	0.030	0.100	0.088	0.058	0.067
EATTA-style	label queries	10.0	0.120	0.050	0.000	0.148	0.094	0.000
Full-label active	label queries	10.0	0.073	0.031	0.033	0.089	0.058	0.000

- REPRODUCE.md
- RUN_LOG.md
- RESULTS_SUMMARY.md
- FINAL_REPORT_STAGE2.md
- audit/repro_audit.md
- audit/missing_outputs.md

The principal result files supporting the PPSN analyses are:

- results/raw/frontier_stage2_final_cifar10/frontier_runs.csv
- results/raw/contamination_stage2_final_cifar10/contamination_runs.csv
- results/raw/frontier_stage2_final_cifar100/frontier_runs.csv
- results/raw/contamination_stage2_final_cifar100/contamination_runs.csv
- results/raw/robustness_noise_stage2_cifar10/robustness_runs.csv
- results/raw/robustness_delay_stage2_cifar10/robustness_runs.csv

Table 3. Full stationary frontier table at $q = 0.01$ with paired bootstrap intervals for gain over frozen.

Dataset	Method	ID acc	Gain vs frozen	NTR	ECE
CIFAR-10-C	Frozen	0.623 +/- 0.174	0.000 [0.000, 0.000]	0.000	0.244
CIFAR-10-C	Tent	0.775 +/- 0.079	0.151 [0.125, 0.181]	0.000	0.130
CIFAR-10-C	FB-Gate-Tent	0.770 +/- 0.081	0.147 [0.121, 0.175]	0.000	0.132
CIFAR-10-C	Self-Gate Tent (0-bit)	0.773 +/- 0.080	0.149 [0.123, 0.178]	0.000	0.130
CIFAR-10-C	Throttle Tent (0-bit)	0.717 +/- 0.102	0.094 [0.068, 0.120]	0.000	0.170
CIFAR-10-C	COME-style 0-bit	0.773 +/- 0.080	0.150 [0.122, 0.177]	0.000	0.126
CIFAR-10-C	BiTTA-style	0.714 +/- 0.124	0.090 [0.075, 0.105]	0.000	0.186
CIFAR-10-C	ATTA-style	0.694 +/- 0.138	0.071 [0.061, 0.082]	0.000	0.196
CIFAR-10-C	EATTA-style	0.743 +/- 0.107	0.120 [0.100, 0.140]	0.000	0.159
CIFAR-10-C	Full-label active	0.696 +/- 0.138	0.073 [0.063, 0.084]	0.000	0.196
CIFAR-100-C	Frozen	0.374 +/- 0.164	0.000 [0.000, 0.000]	0.000	0.317
CIFAR-100-C	Tent	0.553 +/- 0.093	0.179 [0.156, 0.203]	0.000	0.202
CIFAR-100-C	FB-Gate-Tent	0.530 +/- 0.108	0.156 [0.134, 0.178]	0.000	0.214
CIFAR-100-C	Self-Gate Tent (0-bit)	0.542 +/- 0.094	0.167 [0.143, 0.190]	0.000	0.206
CIFAR-100-C	Throttle Tent (0-bit)	0.553 +/- 0.093	0.179 [0.156, 0.202]	0.000	0.202
CIFAR-100-C	COME-style 0-bit	0.552 +/- 0.093	0.178 [0.155, 0.200]	0.000	0.193
CIFAR-100-C	BiTTA-style	0.475 +/- 0.119	0.101 [0.086, 0.114]	0.000	0.275
CIFAR-100-C	ATTA-style	0.463 +/- 0.129	0.088 [0.078, 0.099]	0.000	0.268
CIFAR-100-C	EATTA-style	0.523 +/- 0.100	0.148 [0.129, 0.168]	0.000	0.228
CIFAR-100-C	Full-label active	0.463 +/- 0.131	0.089 [0.078, 0.099]	0.000	0.272

- `results/raw/robustness_noise_stage2_cifar100/robustness_runs.csv`
- `results/raw/robustness_delay_stage2_cifar100/robustness_runs.csv`

Table 4. Full contamination table at $q = 0.01$ and $r = 0.5$ with paired bootstrap intervals for gain over frozen.

Dataset	Method	ID acc	Gain vs frozen	NTR	Accept
CIFAR-10-C + SVHN	Frozen	0.743 +/- 0.082	0.000 [0.000, 0.000]	0.000	0.000
CIFAR-10-C + SVHN	Tent	0.788 +/- 0.074	0.045 [0.031, 0.059]	0.133	1.000
CIFAR-10-C + SVHN	FB-Gate-Tent	0.773 +/- 0.076	0.029 [0.020, 0.040]	0.000	0.163
CIFAR-10-C + SVHN	Self-Gate Tent (0-bit)	0.792 +/- 0.074	0.048 [0.036, 0.061]	0.100	0.443
CIFAR-10-C + SVHN	Throttle Tent (0-bit)	0.788 +/- 0.074	0.045 [0.032, 0.059]	0.133	0.963
CIFAR-10-C + SVHN	COME-style 0-bit	0.793 +/- 0.072	0.049 [0.036, 0.064]	0.133	1.000
CIFAR-10-C + SVHN	BiTTA-style	0.751 +/- 0.093	0.007 [-0.004, 0.018]	0.400	1.000
CIFAR-10-C + SVHN	ATTA-style	0.773 +/- 0.058	0.030 [0.020, 0.040]	0.100	0.923
CIFAR-10-C + SVHN	EATTA-style	0.793 +/- 0.065	0.050 [0.041, 0.059]	0.000	0.607
CIFAR-10-C + SVHN	Full-label active	0.774 +/- 0.062	0.031 [0.022, 0.040]	0.033	0.620
CIFAR-100-C + SVHN	Frozen	0.467 +/- 0.129	0.000 [0.000, 0.000]	0.000	0.000
CIFAR-100-C + SVHN	Tent	0.540 +/- 0.101	0.073 [0.056, 0.089]	0.067	1.000
CIFAR-100-C + SVHN	FB-Gate-Tent	0.514 +/- 0.110	0.047 [0.036, 0.058]	0.000	0.170
CIFAR-100-C + SVHN	Self-Gate Tent (0-bit)	0.536 +/- 0.106	0.069 [0.055, 0.084]	0.033	0.327
CIFAR-100-C + SVHN	Throttle Tent (0-bit)	0.540 +/- 0.101	0.073 [0.057, 0.090]	0.067	1.000
CIFAR-100-C + SVHN	COME-style 0-bit	0.545 +/- 0.101	0.077 [0.060, 0.094]	0.033	1.000
CIFAR-100-C + SVHN	BiTTA-style	0.493 +/- 0.115	0.026 [0.014, 0.037]	0.167	1.000
CIFAR-100-C + SVHN	ATTA-style	0.525 +/- 0.092	0.058 [0.042, 0.073]	0.067	0.990
CIFAR-100-C + SVHN	EATTA-style	0.561 +/- 0.080	0.094 [0.074, 0.113]	0.000	1.000
CIFAR-100-C + SVHN	Full-label active	0.525 +/- 0.090	0.058 [0.042, 0.073]	0.000	0.917

Table 5. Primary robustness table for noise and delayed commitment at $q = 0.01$ and $r = 0.5$.

Dataset	Axis	Value	ID acc	Gain	NTR	Accept
CIFAR-10-C + SVHN	Noise eta	0	0.773	0.029	0.000	0.163
CIFAR-10-C + SVHN	Noise eta	0.05	0.775	0.032	0.033	0.323
CIFAR-10-C + SVHN	Noise eta	0.1	0.774	0.031	0.033	0.360
CIFAR-10-C + SVHN	Noise eta	0.2	0.779	0.035	0.033	0.303
CIFAR-10-C + SVHN	Delay d	0	0.773	0.029	0.000	0.163
CIFAR-10-C + SVHN	Delay d	1	0.769	0.025	0.000	0.163
CIFAR-10-C + SVHN	Delay d	5	0.746	0.003	0.000	0.233
CIFAR-100-C + SVHN	Noise eta	0	0.514	0.047	0.000	0.170
CIFAR-100-C + SVHN	Noise eta	0.05	0.517	0.050	0.000	0.333
CIFAR-100-C + SVHN	Noise eta	0.1	0.520	0.053	0.000	0.407
CIFAR-100-C + SVHN	Noise eta	0.2	0.522	0.055	0.000	0.340
CIFAR-100-C + SVHN	Delay d	0	0.514	0.047	0.000	0.170
CIFAR-100-C + SVHN	Delay d	1	0.507	0.040	0.000	0.170
CIFAR-100-C + SVHN	Delay d	5	0.476	0.009	0.000	0.317

Table 6. Compute and reproducibility table derived from the recorded execution history.

Study	Runs	Runtime / stream (s)	Peak memory (MB)	Raw path
C10 frontier final	4200	1.039	1200.8	`results/raw/frontier_stage2_final_cifar10/frontier_runs.csv`
C10 contamination final	10500	1.180	1196.4	`results/raw/contamination_stage2_final_cifar10/contamination_runs.csv`
C100 frontier final	4200	0.975	1236.8	`results/raw/frontier_stage2_final_cifar100/frontier_runs.csv`
C100 contamination final	10500	0.985	1229.7	`results/raw/contamination_stage2_final_cifar100/contamination_runs.csv`
C10 robustness noise	8400	1.017	1086.5	`results/raw/robustness_noise_stage2_cifar10/robustness_runs.csv`
C10 robustness delay	6300	0.928	1086.5	`results/raw/robustness_delay_stage2_cifar10/robustness_runs.csv`
C100 robustness noise	8400	0.928	1087.1	`results/raw/robustness_noise_stage2_cifar100/robustness_runs.csv`
C100 robustness delay	6300	0.825	1087.2	`results/raw/robustness_delay_stage2_cifar100/robustness_runs.csv`
Source train cifar100	3	173.765	1372.8	`results/raw/source_train/cifar100`